PROFESSIONAL PORTFOLIO

Godwin Etim Akpan

Data Analytics | Data Engineering | Enterprise GIS | GeoAI | Machine Learning

Selected projects in spatial intelligence, public-health analytics, secure data systems, climate and hazard exposure, and operational decision support

10+ YEARS	6,000+	100+	10+
GIS, analytics and spatial decision support	Maps, dashboards and analytical products	Professionals trained	Peer-reviewed publications

Portfolio Website • LinkedIn • godwinea.ai@gmail.com

Raleigh, North Carolina | July 2026

PROFESSIONAL PROFILE

Turning complex data into trusted intelligence

I am a GIS, data analytics and data engineering professional with more than 10 years of experience transforming spatial, public-health and enterprise data into practical intelligence for informed decision-making.

My work combines enterprise GIS, spatial analysis, data engineering, GeoAI, machine learning, cloud computing, explainable AI and secure software engineering. I develop scalable, governance-aware systems supporting public health, emergency management, humanitarian response, infrastructure planning and organisational decision-making.

Professional evidence

- National-level public-health surveillance and emergency-response support in Liberia.
- Applied outbreak intelligence spanning COVID-19, malaria, measles, cholera, mpox, Ebola and Lassa fever.
- Enterprise GIS implementation, spatial data infrastructure, dashboard development and operational reporting.
- GIS training, capacity building, data-quality assurance, automation and applied spatial epidemiology.

CORE CAPABILITIES

Data Analytics

Python, SQL, dashboards, predictive analytics and business intelligence

Data Engineering

ETL/ELT, validation, feature engineering, PostgreSQL/PostGIS, Spark and PySpark

Enterprise GIS

ArcGIS Pro/Online, QGIS, spatial infrastructure, cartography and remote sensing

GeoAI & Machine Learning

XGBoost, Scikit-learn, spatial statistics, SHAP and responsible AI

Secure Systems

FastAPI, Docker, AWS, JWT, RBAC, audit logging and privacy-aware analytics

ACADEMIC FOUNDATION

MSc Big Data Technologies — Candidate

University of East London / UNICAF · Final award pending

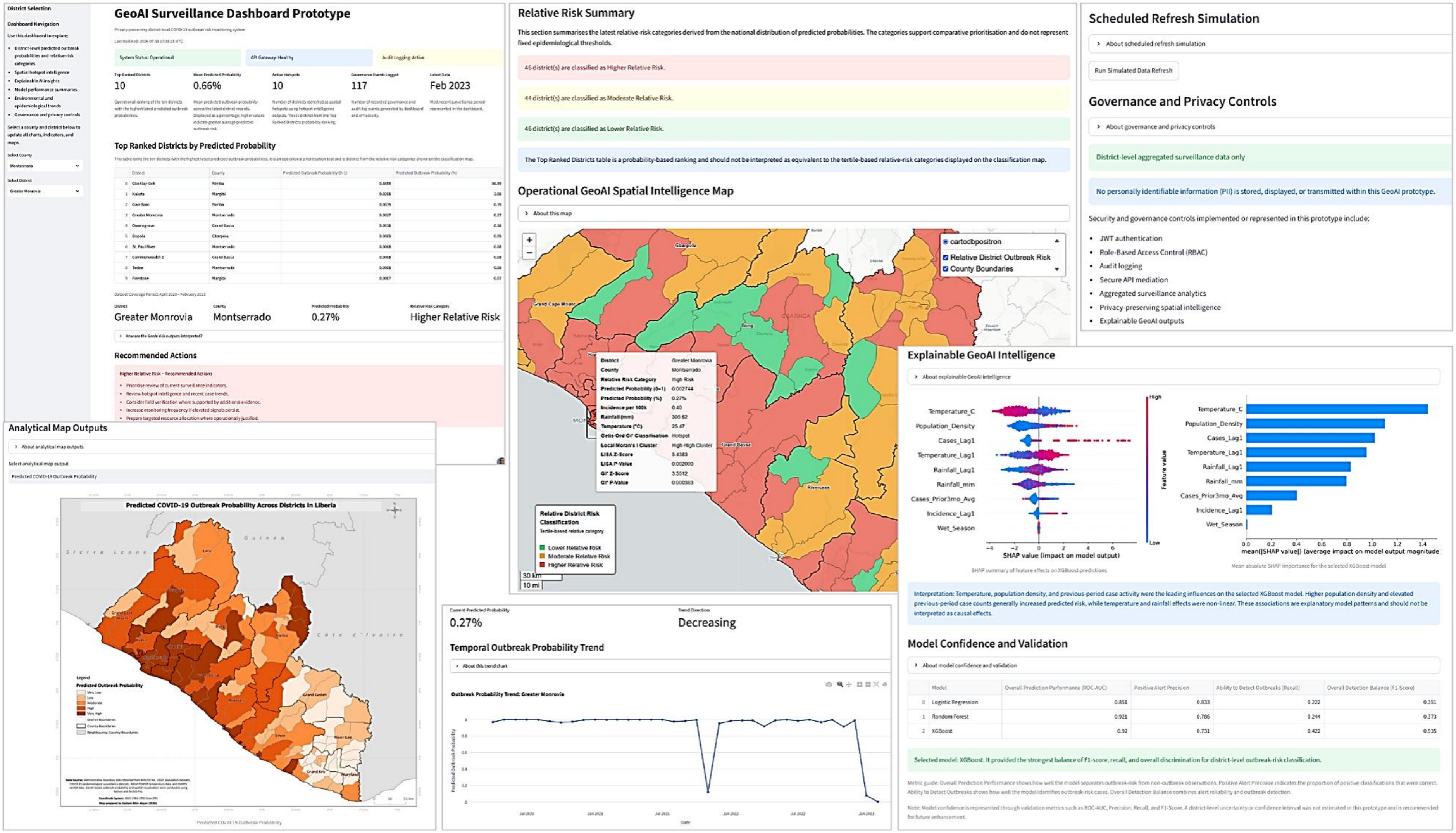
M.Tech. Geoinformation Science and Remote Sensing

FUTA, Nigeria / UN-ARCSSTEE

FLAGSHIP PROJECT

Privacy-Preserving GeoAI Health Surveillance Dashboard

An integrated research prototype for risk ranking, spatial intelligence, explainability, governance monitoring and decision support.

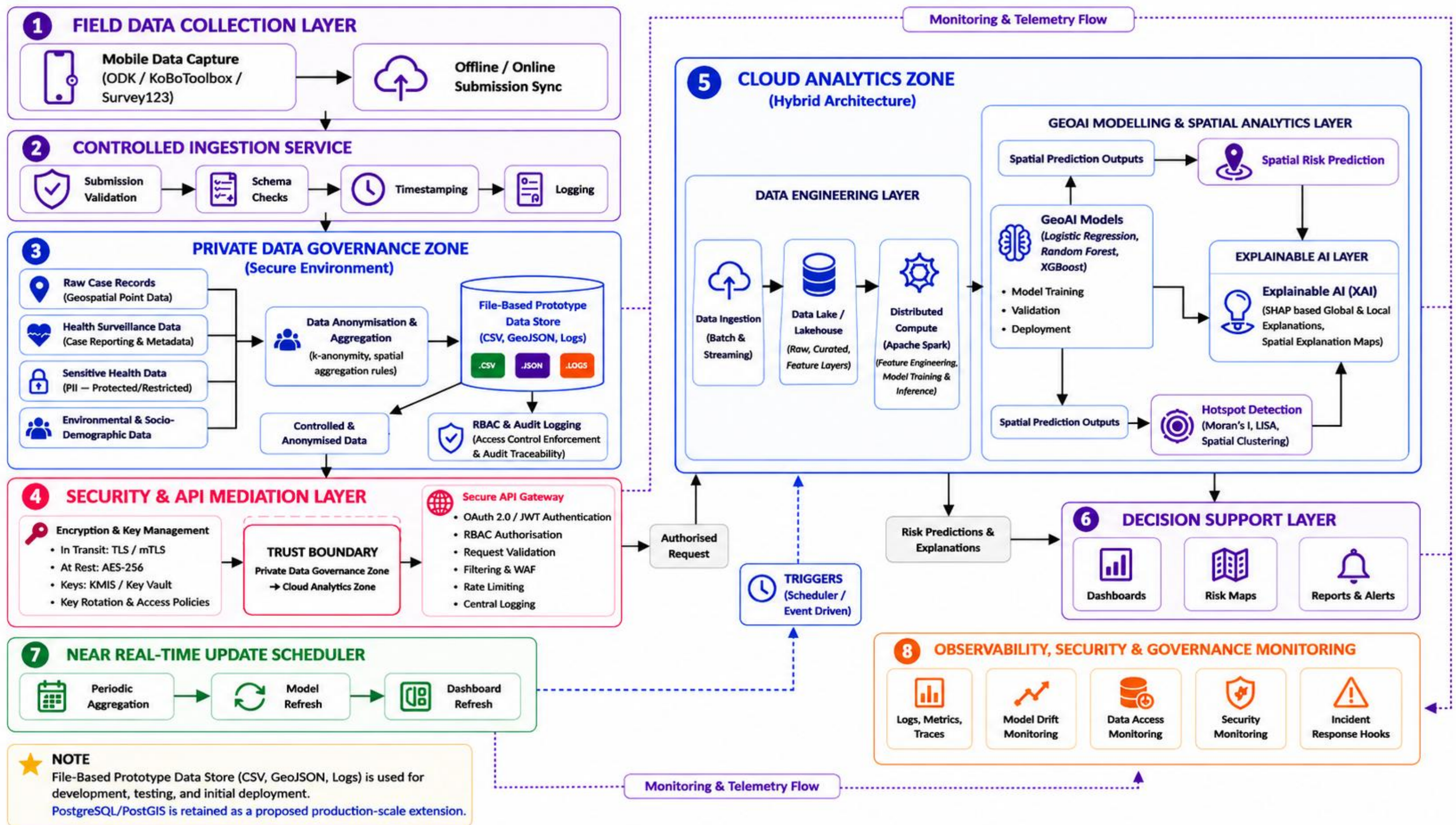


Integrated dashboard showing risk rankings, interactive mapping, SHAP explanations, governance controls and model validation.

Research-prototype boundary: The public demonstration uses aggregated, de-identified district-level data and is not intended for clinical diagnosis or operational deployment without further validation, security hardening, governance review and institutional approval.

End-to-End Privacy-Preserving GeoAI Architecture

A production-oriented reference architecture connecting controlled data ingestion, governance, secure API mediation, GeoAI analytics and monitored decision support.

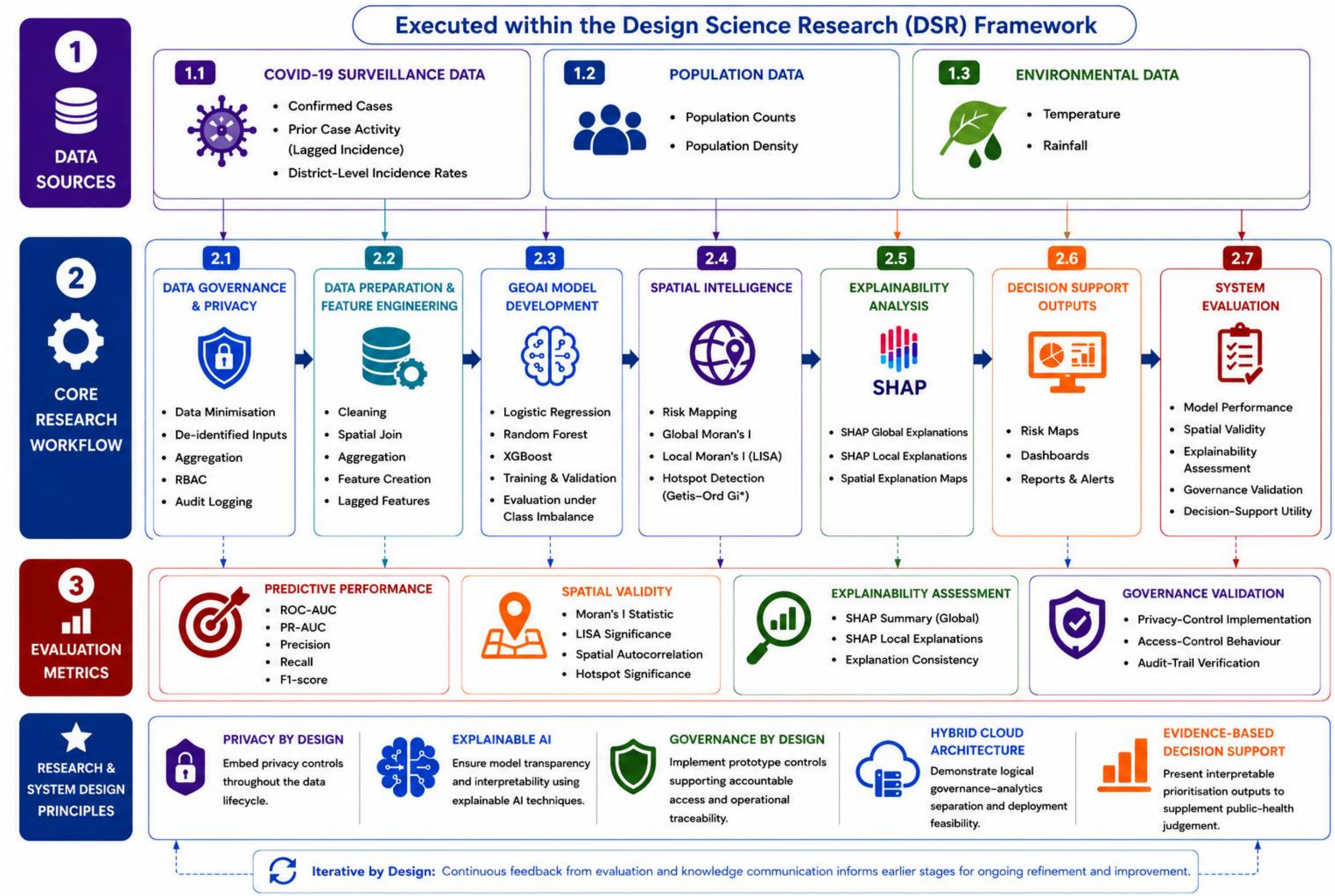


Reference architecture developed for the dissertation research artefact.

Implementation scope: The prototype implemented the core data preparation, spatial analytics, model evaluation, dashboard, FastAPI gateway, JWT authentication, RBAC and audit-logging functions. Spark/lakehouse, WAF, key-management and advanced observability elements represent production-scale extensions.

Design Science Research Workflow

The dissertation evaluates the artefact as an integrated research and engineering system—not simply as a predictive model.



Research workflow spanning data sources, governance, feature engineering, GeoAI, spatial intelligence, explainability and evaluation.

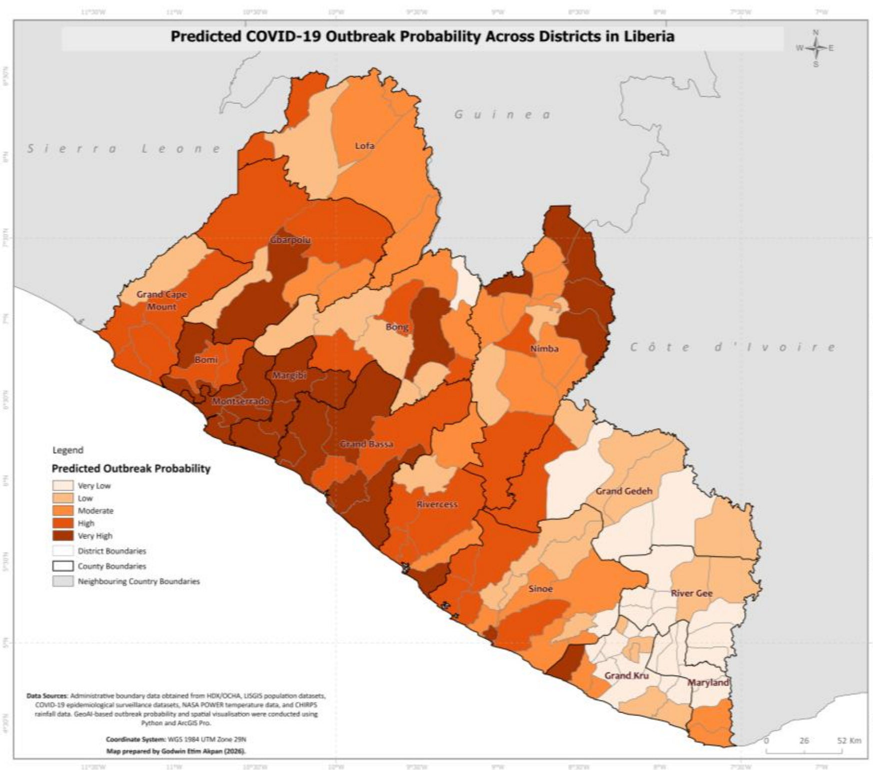
Evaluation dimensions: Predictive performance · spatial validity · explainability · security and governance · usability · decision-support value.

GEOAI OUTPUTS

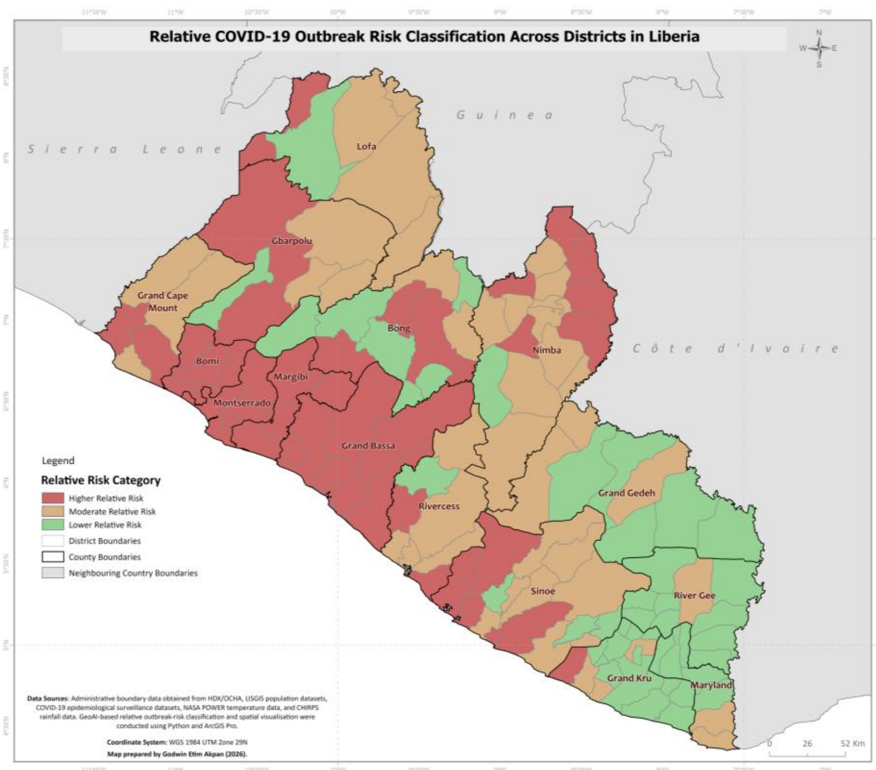
From Predicted Probability to Decision-Support Classification

Probability supports analytical ranking; relative-risk categories provide an interpretable comparative view for prioritisation.

Predicted outbreak probability



Relative district-risk category



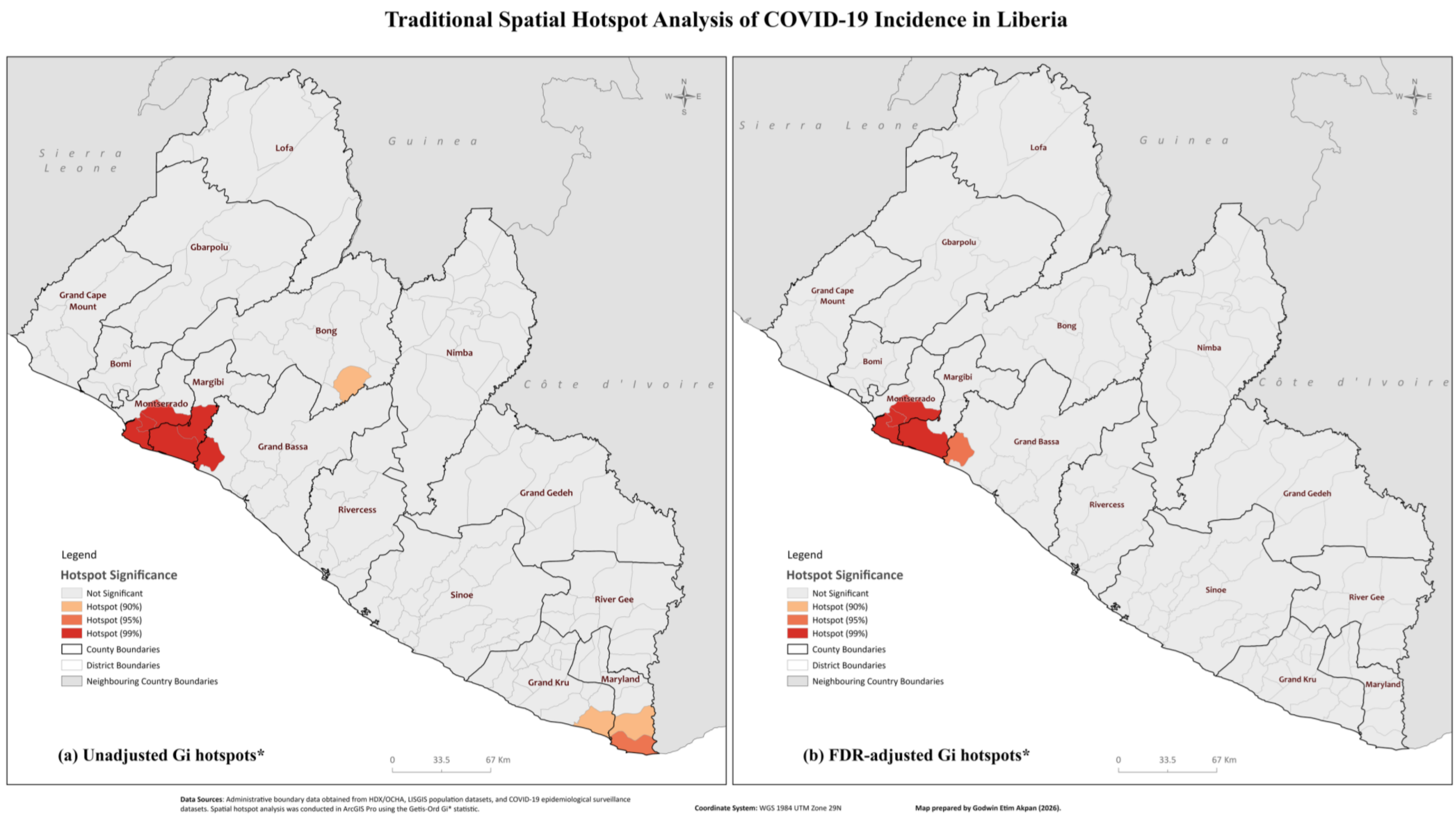
District-level demonstration outputs derived from the dissertation's XGBoost-based GeoAI workflow.

Interpretation: The maps communicate comparative risk and model output; they do not represent fixed epidemiological thresholds or clinical predictions.

SPATIAL STATISTICS

Hotspot Sensitivity and Multiple-Testing Adjustment

Comparing original Getis-Ord Gi* classifications with false-discovery-rate-adjusted results supports more defensible spatial interpretation.



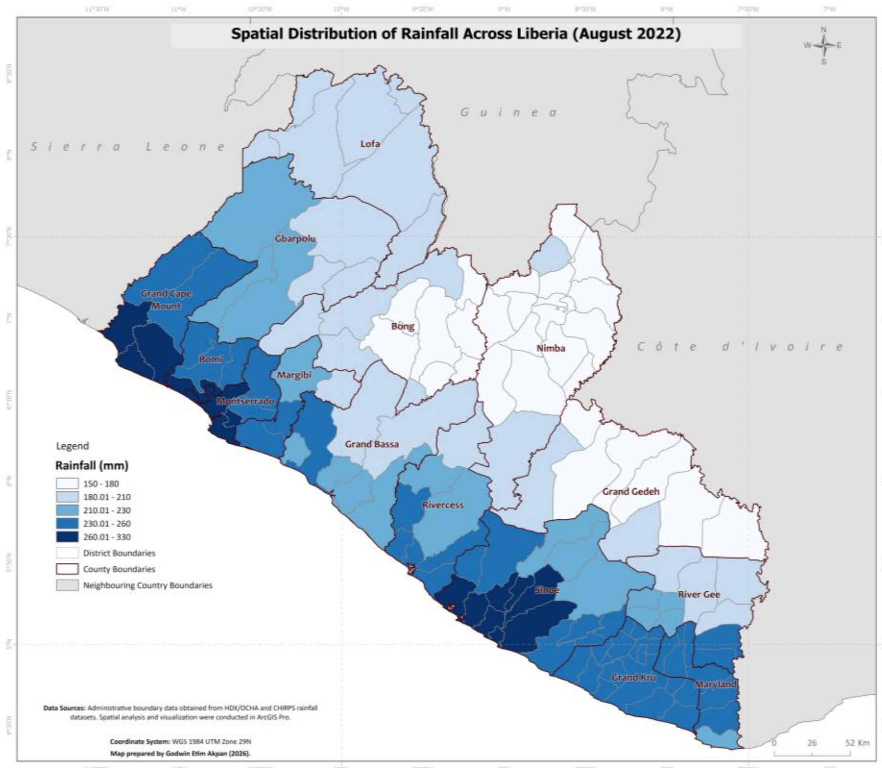
Traditional hotspot analysis of COVID-19 incidence: unadjusted Gi results compared with FDR-adjusted results.*

Analytical value: This comparison demonstrates awareness that apparently significant spatial patterns can change after correction for multiple testing—an important safeguard against overinterpretation.

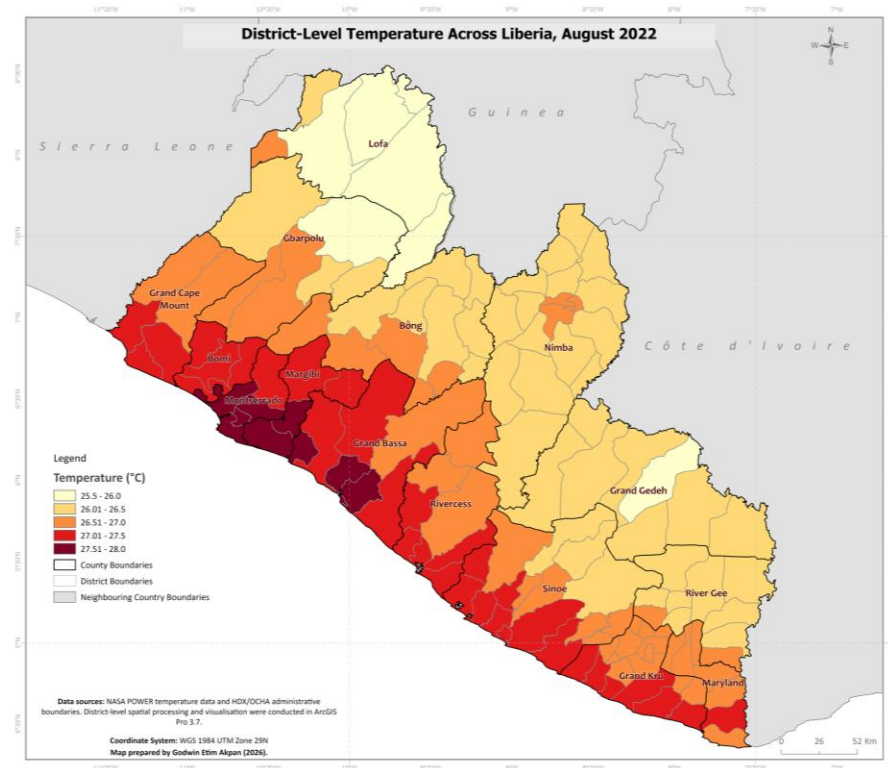
Rainfall and Temperature as Spatial Risk Features

Environmental indicators were spatially joined to district-level surveillance and population data for exploratory modelling and explanation.

Rainfall distribution — August 2022



District-level temperature — August 2022



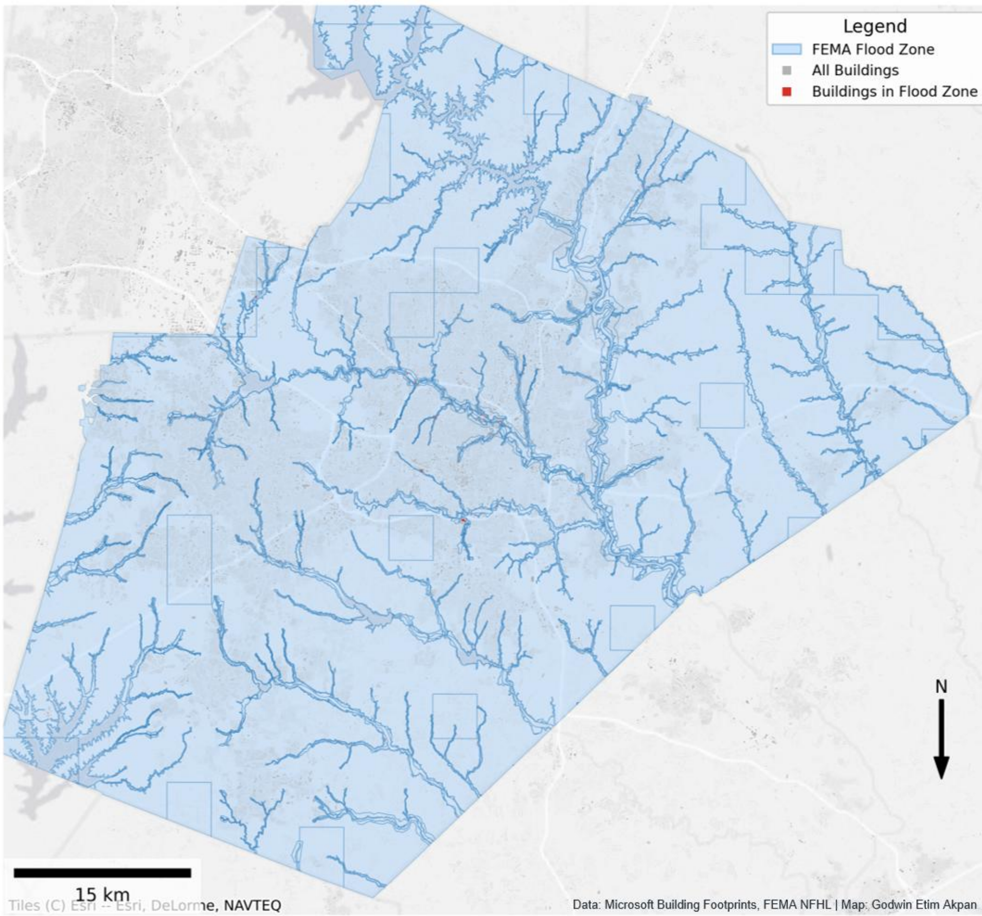
Environmental data: CHIRPS rainfall and NASA POWER temperature; administrative boundaries from HDX/OCHA and national sources.

NORTH CAROLINA HAZARD EXPOSURE

Wake County Building Exposure in FEMA Flood Zones

A reproducible portfolio demonstration using public FEMA flood-hazard data and Microsoft building footprints.

Wake County, NC — Building Exposure in FEMA Flood Zones
Demo Analysis — Not for Operational Use



PURPOSE

Identify building footprints intersecting mapped FEMA flood zones and demonstrate an emergency-management and infrastructure-exposure workflow.

METHOD

- Acquired and validated public building and hazard layers.
- Standardised projection and geometry.
- Applied spatial overlay and intersection analysis.
- Classified exposed features and produced a decision-support map.

PORTFOLIO BOUNDARY

Demonstration analysis only. It was not commissioned by NCEM and is not intended for operational use.

Recommended next step

Add an enlarged inset, exposure counts and flood-zone breakdown in the next analytical iteration.

OPERATIONAL DELIVERY

Emergency GIS Data Engineering and QA/QC

Trusted decision-support products begin with controlled ingestion, validated attributes, spatial consistency and reproducible publishing.

1	2	3	4	5
UNDERSTAND	**PREPARE**	**INTEGRATE**	**DELIVER**	**SUSTAIN**
Clarify operational users, decisions, timing, data owners and minimum viable outputs.	Validate schema, clean attributes, standardise coordinate systems and resolve geometry issues.	Join spatial and non-spatial data, engineer features and document lineage and assumptions.	Publish maps, dashboards, reports and APIs with clear interpretation and responsible-use notes.	Monitor refreshes, troubleshoot data quality, maintain documentation and support users.

Selected evidence

- Digitised and validated Liberia's road network using field-collected GPS observations and satellite imagery.
- Standardised road class, condition and accessibility attributes for integration into national GIS datasets.
- Applied validation, cleaning, aggregation, feature engineering and spatial joins to prepare model-ready surveillance data.
- Delivered time-sensitive GIS products supporting outbreak response, planning and operational reporting.

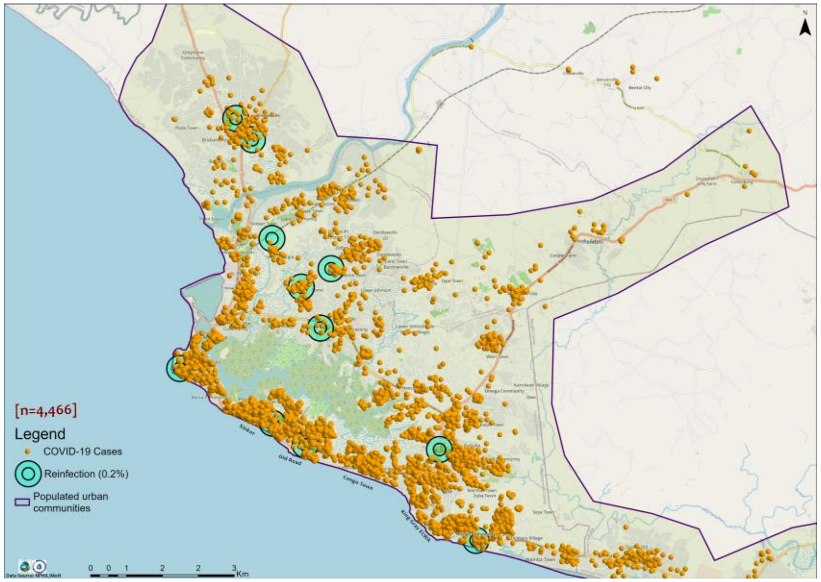
Engineering stack

Python · SQL · Pandas · GeoPandas · PostgreSQL/PostGIS · ETL/ELT · data validation · feature engineering · ArcGIS Pro · QGIS · FastAPI · Docker · Git/GitHub

Delivery principle

Every analytical result should be traceable to its data, assumptions, validation checks and intended decision context.

Emergency Operations and Time-Series Public Health Surveillance

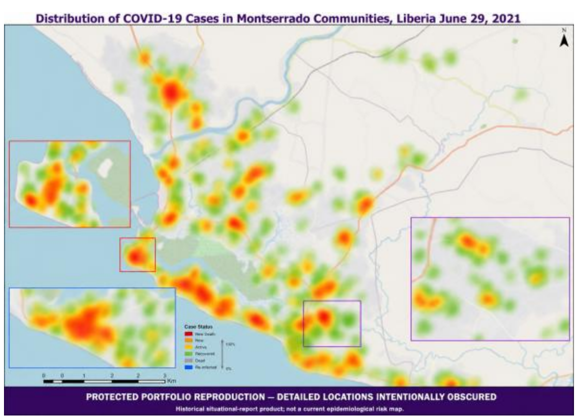


Published spatial epidemiology research: Spatial distribution and clustering of COVID-19 cases in Liberia. Reproduced from Akpan et al. (2022), *COVID-19 Reinfection in Liberia: Implication for Improving Disease Surveillance*, PLOS ONE, 17(3), e0265768. https://doi.org/10.1371/journal.pone.0265768. Licensed under CC BY 4.0.

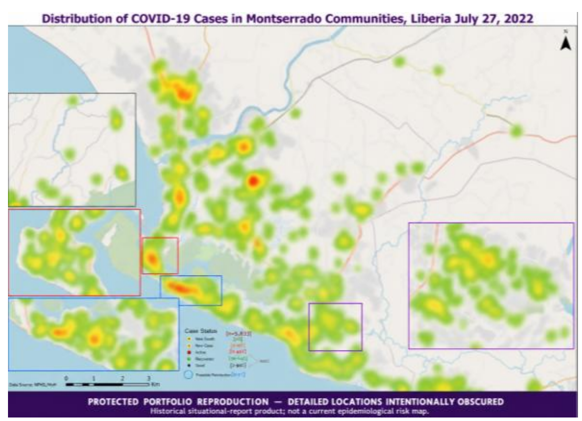
Supported national Ebola and COVID-19 emergency-response operations by producing daily and on-demand GIS products that integrated evolving surveillance and geographic information for situational awareness, resource prioritisation and partner coordination.

- Delivered more than 6,000 maps, dashboards and analytical products.
- Supported field data collection, troubleshooting, reporting and stakeholder communication.
- Trained more than 100 professionals in GIS, spatial analysis and digital data collection.
- Developed validated time-series workflows comparing surveillance patterns across June 2021, July 2022 and February 2023.
- Applied consistent symbology, spatial validation and descriptive analysis without presenting causal or formal hotspot conclusions.

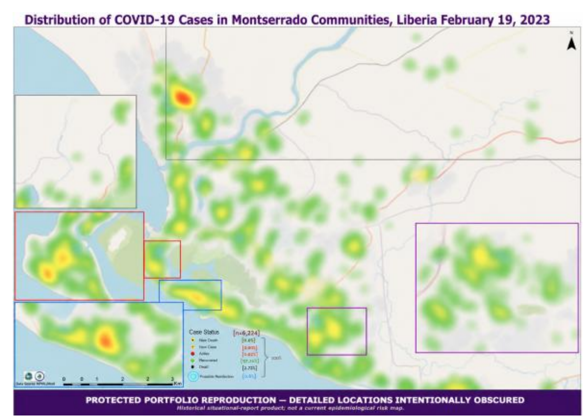
COVID-19 Case Distribution — June 2021



COVID-19 Case Distribution — July 2022



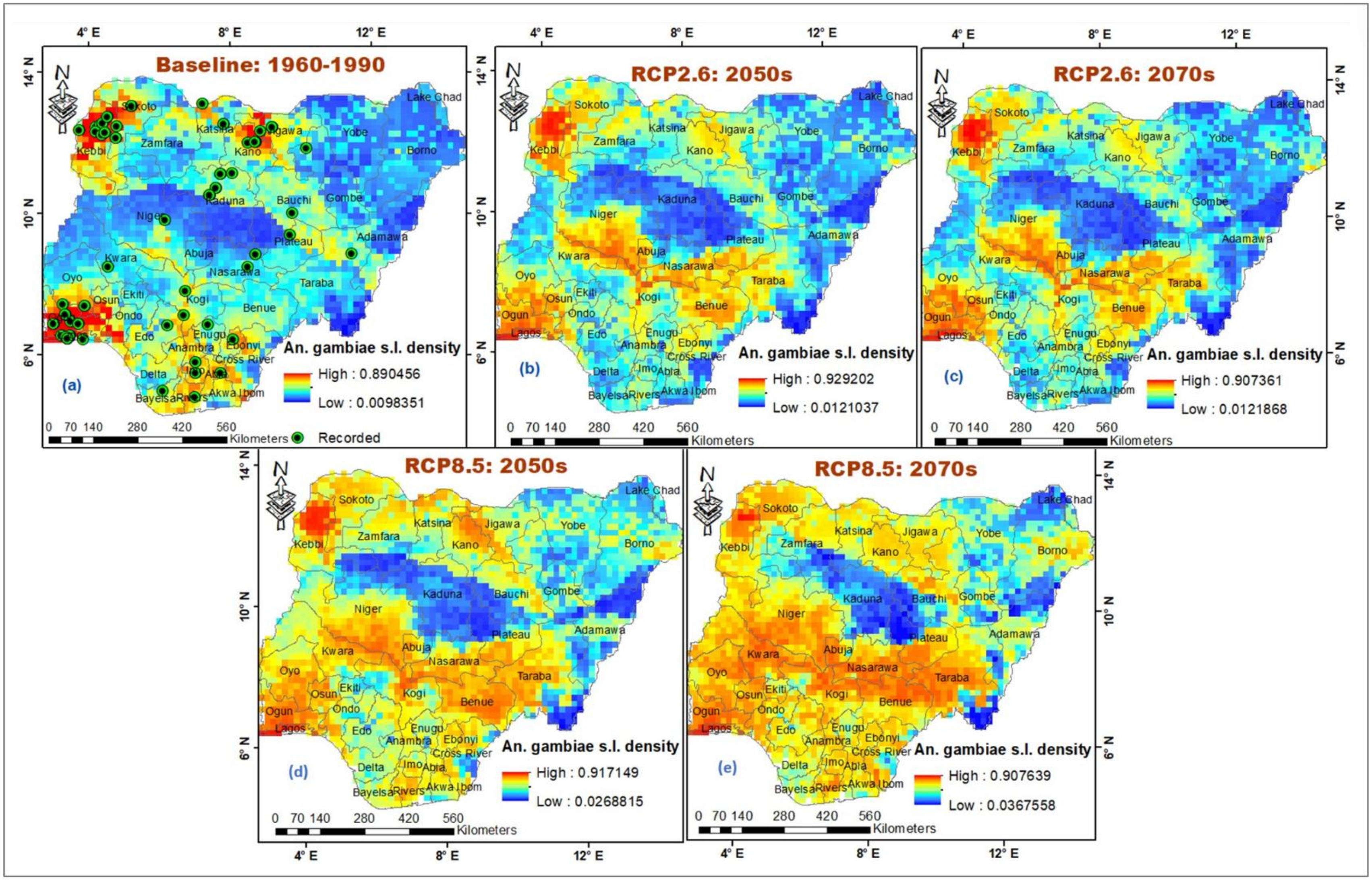
COVID-19 Case Distribution — February 2023



Responsible portfolio presentation: These historical operational maps supported national COVID-19 situational reporting and were previously disseminated through authorised channels. Community labels and precise record locations have been intentionally obscured. The maps demonstrate time-series workflow and operational cartographic capability; they do not represent current epidemiological risk.

Potential Distribution of Malaria Vectors Under Climate Scenarios

A MaxEnt-based spatial modelling project examining how environmental suitability for Anopheles mosquitoes could shift under projected climate conditions in Nigeria.



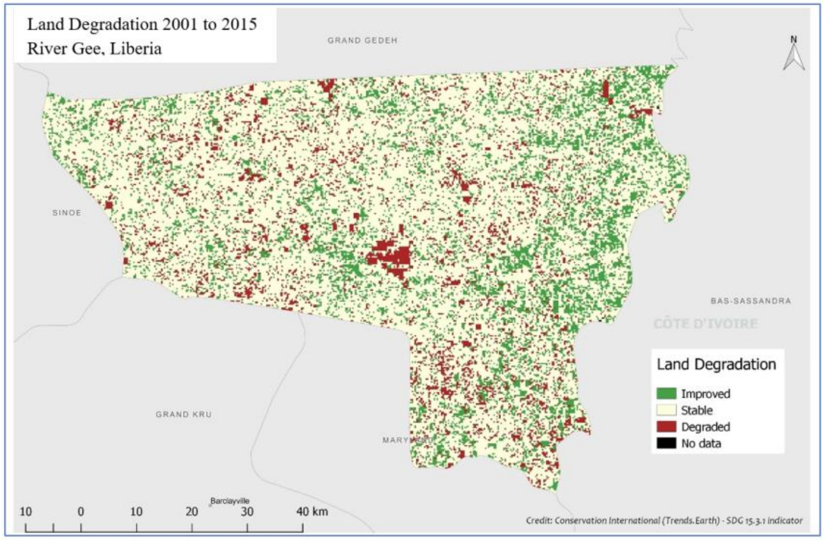
Baseline and projected suitability scenarios produced using MaxEnt and GIS-based bioclimatic modelling.

Capabilities demonstrated: Species-distribution modelling · remote sensing · climate scenarios · environmental raster processing · spatial interpretation.

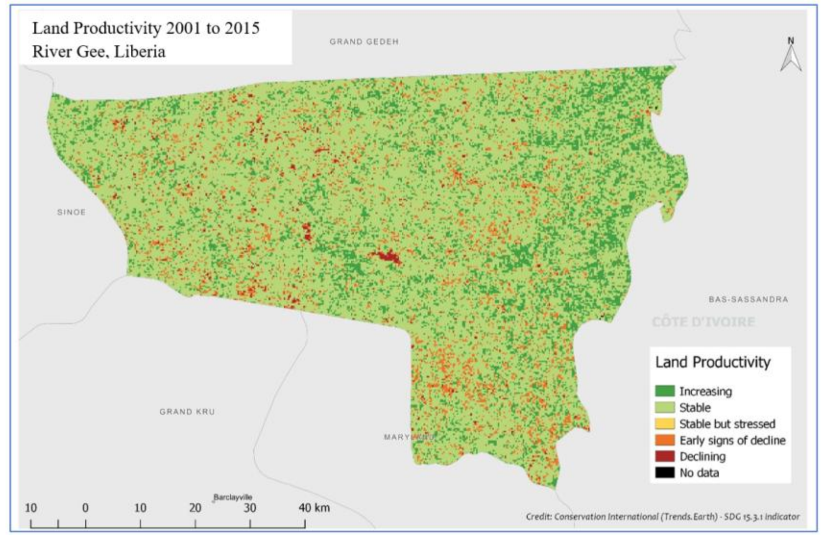
Land Degradation and Productivity Assessment — River Gee County

An SDG 15.3 pilot project assessing land-productivity trends, land-cover dynamics and environmental change from 2001 to 2015.

Land degradation



Land productivity



PROJECT CONTRIBUTION

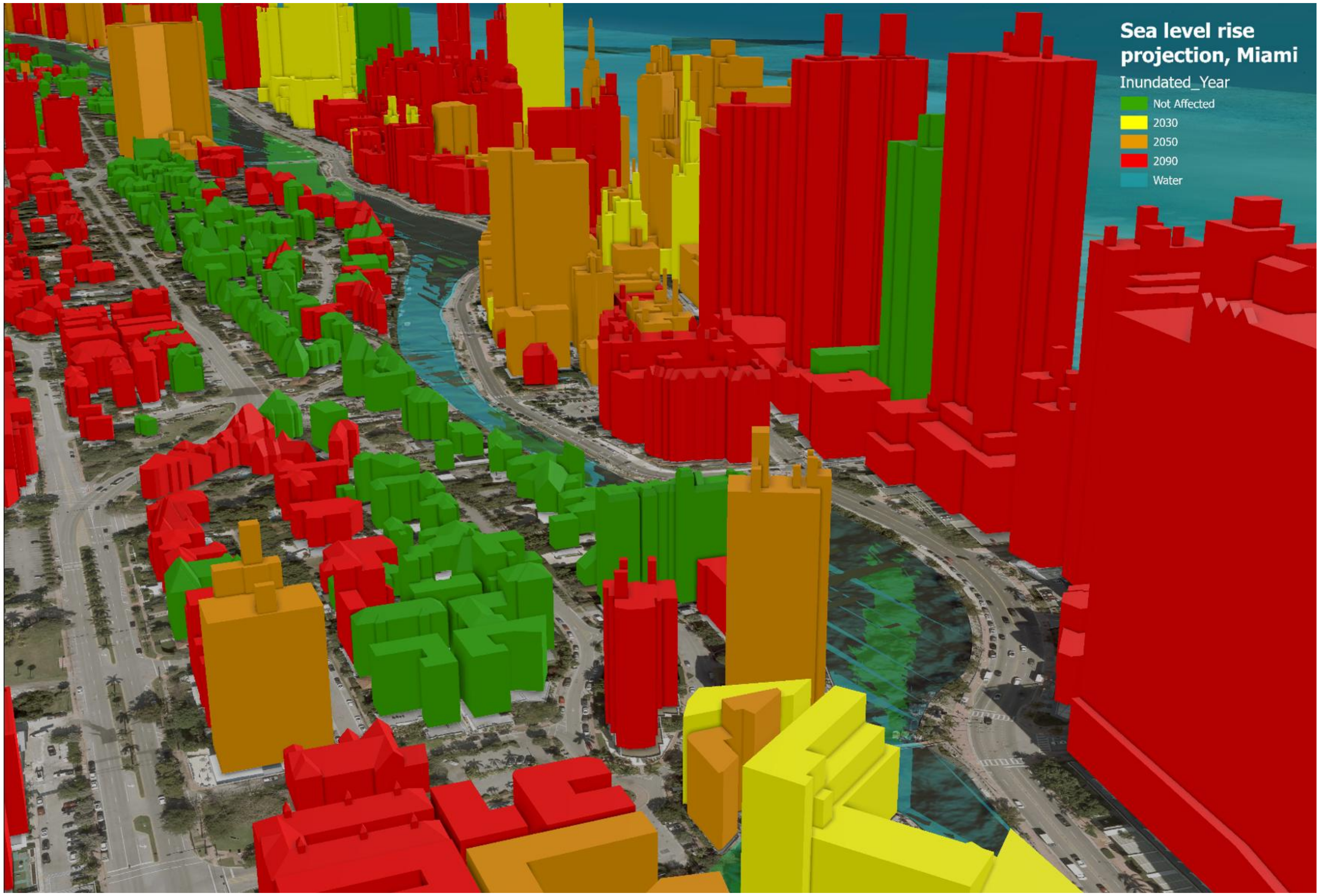
- Designed and executed an independent SDG-monitoring proof of concept.
- Processed productivity, land-cover and soil-organic-carbon indicators.
- Mapped land-degradation and productivity patterns at high resolution.
- Translated technical results into stakeholder-oriented environmental intelligence.

TOOLS

Trends.Earth · ArcGIS · remote sensing · spatial analysis

Sea-Level-Rise Projection — Miami, Florida

A time-enabled 3D visualisation of climate-induced exposure designed to communicate changing urban vulnerability.



3D building visualisation using elevation, sea-level-rise scenarios and time-based symbology.

Analysis

Elevation and land-cover data prepared for scenario-based visualisation.

Communication

3D extrusion and colour communicate relative exposure and changing risk.

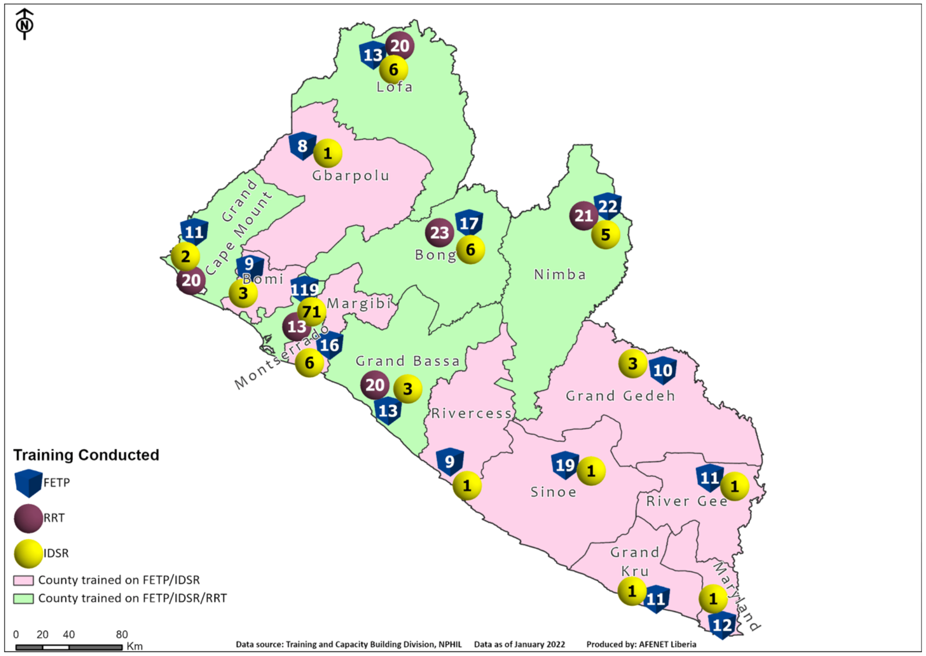
Application

Supports climate resilience, planning discussions and public-facing risk communication.

Workforce Coverage and Cross-Border Surveillance

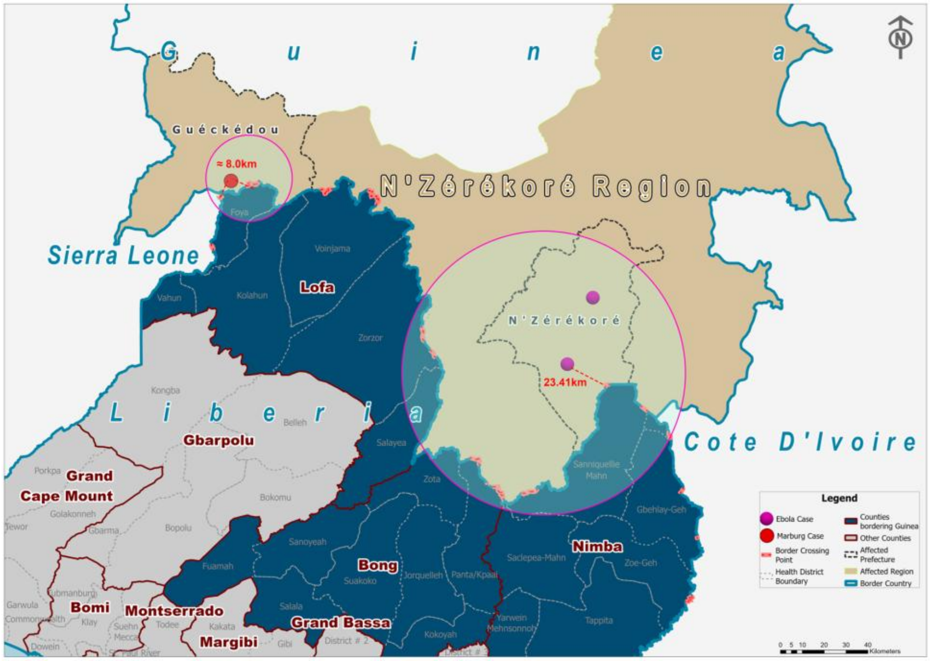
Operational cartography supporting public-health capacity building, incident preparedness and geographically targeted surveillance.

Emergency preparedness training coverage



Visualises national workforce-development reach and identifies counties with limited or no documented coverage.

Ebola and Marburg cross-border mapping



Integrates incident proximity, administrative boundaries and border context to support preparedness and surveillance prioritisation.

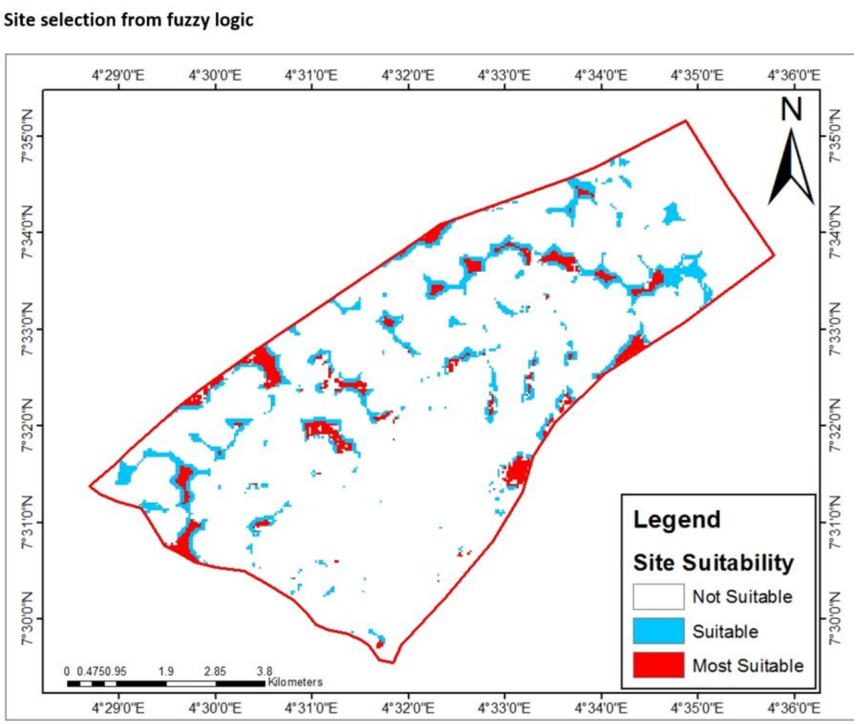
Selected operational public-health mapping undertaken in Liberia with institutional partners.

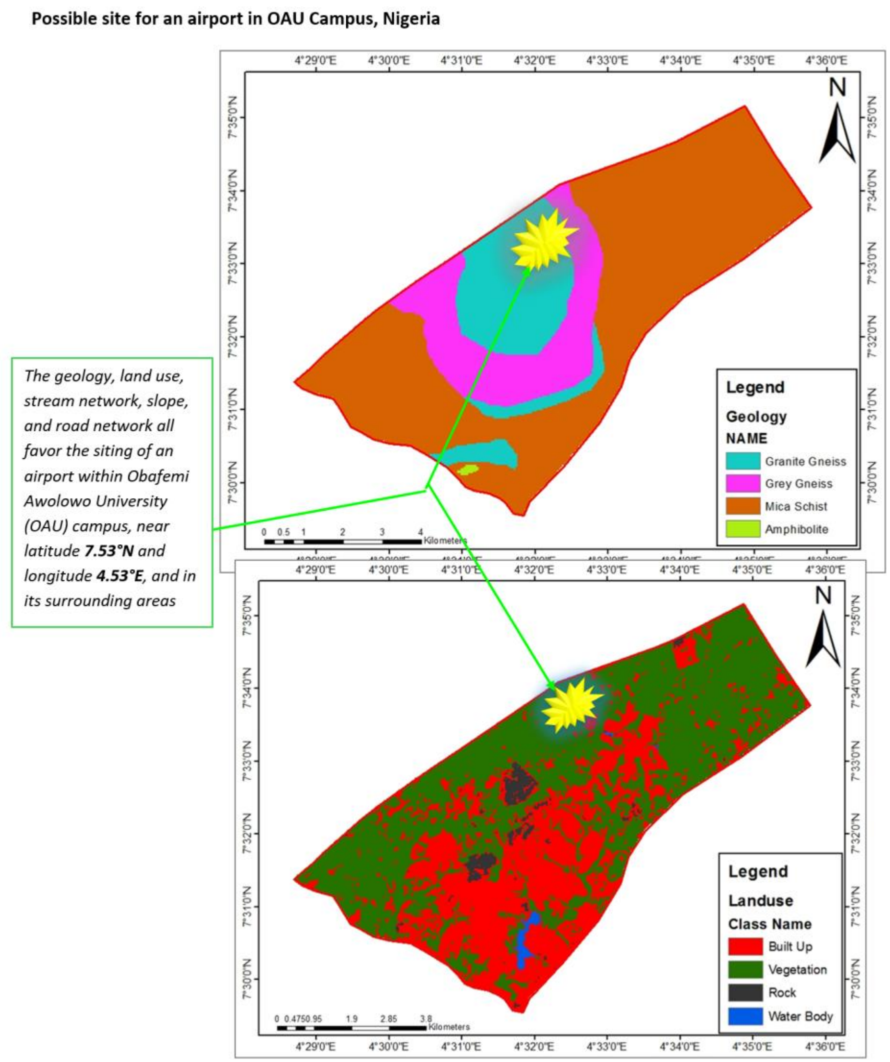
Publication note: Only aggregated or publicly releasable content should be included in externally distributed versions; organisational attribution and permissions should be retained.

INFRASTRUCTURE PLANNING

Airport Site-Suitability Modelling — Obafemi Awolowo University

A multi-criteria GIS analysis integrating geology, land use, stream networks, terrain slope and road access to identify candidate areas.





Fuzzy-logic suitability surface and candidate-site evaluation produced using ArcGIS, Landsat 8 and SRTM elevation data.

Methods demonstrated: Criteria standardisation · proximity analysis · slope and hydrology constraints · fuzzy overlay · candidate-site screening.

Publications, Learning and Collaboration

Publications and research

I have authored and co-authored 10+ peer-reviewed publications covering infectious-disease surveillance, spatial epidemiology, GIS, GeoAI and public health intelligence.

Google Scholar

ORCID 0000-0001-8204-9219

ResearchGate

Current learning focus

- IBM Data Engineering Professional Certificate
- Microsoft Fabric and modern analytics engineering
- Cloud-native data engineering and lakehouse architectures
- GeoAI, spatial machine learning and GeoFoundation Models
- Spatio-temporal AI, AI agents and cloud-security monitoring

OPEN TO OPPORTUNITIES

Data analytics · data engineering · enterprise GIS · GeoAI · spatial data science · machine learning · public health intelligence · secure decision-support systems

CONNECT

Portfolio

GitHub

LinkedIn

Email

Committed to building secure, explainable and practical data solutions that create measurable impact.